

ElectricalVR Ski: Switching of Ankle Tendon Electrical Stimulation for Dynamic Modulation of Ground Shape Perception

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ABSTRACT

This demonstration introduces ElectricalVR Ski, a VR skiing application using ankle tendon electrical stimulation (TES) that presents various ground shape perceptions, dynamically. The novelty of our dynamic sensory presentation system is its employment of ankle TES, which can be controlled dynamically using a computer and its controller are small and lightweight. To achieve the perception of the ground shape that changes dynamically, we investigated the intervals for the switching ankle TES at which the body tilt sensation could be perceived. Then, we confirmed that the user's body tilt sensation was dynamically modulated even when the ankle TES was switched at intervals of 1.0 second.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction devices—Haptic devices

1 INTRODUCTION

Sensory presentation to the lower limbs is essential for achieving an immersive experience in virtual environments (VEs). In particular, ground shape perception is crucial for understanding the VE. Various techniques have been proposed to modulate ground shape perception. For example, a passive haptic feedback system using small bumps to represent the edge of the stairs [2], and a method using a dynamic and walkable pin-array floor device to simulate various types of elevation shapes [1] are proposed. However, each of these methods has inherent limitations. The passive haptic feedback system can only generate sensations at the specific locations where the passive haptics are placed. Consequently, to generate the sensation properly, the user's walking path must be predefined. The pin-array device requires a substantial amount of space to place it, posing challenges for its practical application.

To address both of these limitations, we propose a sensory presentation technique employing ankle tendon electrical stimulation (TES). Ankle TES is known to generate a body tilt sensation and induce body sway by applying electrical stimulation to the ankle tendons of a standing user [3]. It has been reported that ankle TES can present the body tilt sensation in four directions, front-back, left-right, depending on which tendons are stimulated. Therefore, we hypothesized that ground shapes in VEs could be dynamically represented by switching the stimulation points. To our knowledge, no prior studies have investigated the effect of rapidly switching ankle TES. We investigated the threshold intervals of the switching at which the body tilt sensation could be perceived. Consequently, applying the findings, we developed the demonstration of a VR skiing application that uses dynamically switching ankle TES.

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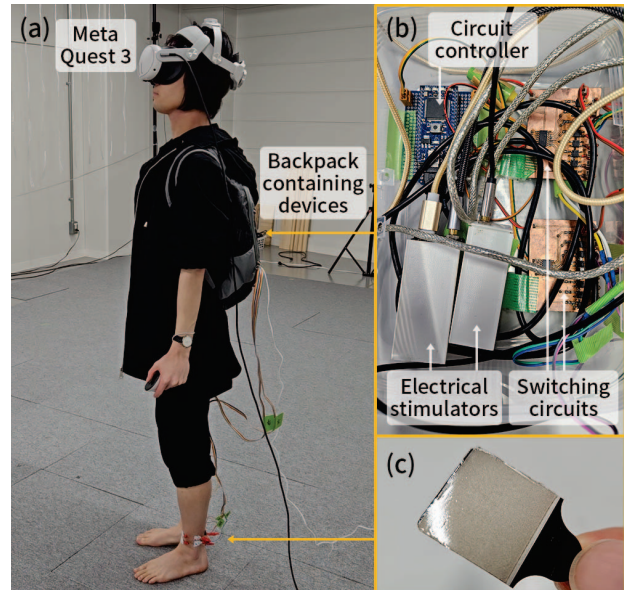


Figure 1: (a): Setup of our demonstration. (b): Electrical stimulators and switching circuit. (c): Electrodes.

2 IMPLEMENTATION

Figure 1 (b) depicts our electrical stimulator. These consist of in-house current control circuits, structured similarly to the electrical stimulator in the previous study [3]. They were designed to be able to deliver an 80-Hz bipolar square current with a maximum of 4 mA. Two electrical stimulators were used to apply current to the ankle tendons of both legs. They are controlled by serial communication signals from a PC. Figure 1 (a) depicts the setup for our demonstration. The electrical stimulators and the switching circuit are stored in a backpack for easy wearing. The electrodes used in this demonstration are 3M Red Dot, featuring an adhesive part measuring approximately 20 mm x 20 mm (Fig. 1 (c)). VR view was provided through the HMD of Meta Quest 3. The user can explore a vast VE by skiing down slopes or being towed by a snowmobile. Figure 2 illustrates the ankle TES conditions in each VR scene. These four conditions of ankle TES are applied depending on the situation in VR, fostering the appropriate ground shape perception that matches visual information. For example, while skiing down the slope, the user is receiving the BS condition's ankle TES, an electrical stimulation that stimulates the Achilles tendons of both feet to induce a forward body tilt sensation. Thus, this system achieved dynamic modulation of ground shape perception with a compact and lightweight setup by switching the stimulation site of the ankle TES.

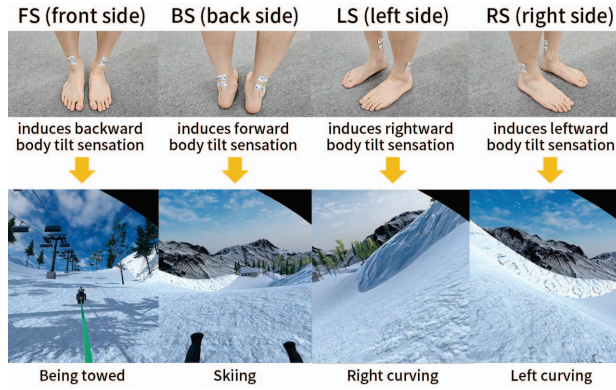


Figure 2: Electrical stimulation conditions applied in each VR scene.

Table 1: The pooled mean of the reaction rate in each switching interval condition.

Switching interval [s]	Reaction rate [%]
0.5	15.3
1.0	94.4
2.0	100.0
4.0	100.0

3 EVALUATION

In this demonstration, ankle TES has to be switched rapidly depending on the user's position in VEs. However, since there is reported latency in the body sway caused by ankle TES, we assumed that if the switching interval is too short, ankle TES may not induce a body tilt sensation effectively. Therefore, we conducted an experiment to investigate the intervals for the switching ankle TES where body tilt sensation is induced. We recruited 8 experimental participants consisting of 7 males and 1 female, with a mean age of 24.2. The participants performed four experimental trials where 2.0 mA of ankle TES was applied for 20 seconds. In each trial, the four electrical stimulation conditions (FS, BS, LS, and RS) were switched at different intervals of 0.5s, 1.0s, 2.0s, and 4.0s. Then, the participants reported the direction of the body tilt sensation in real-time using a dual-axis joystick (KY-023). To control for order effects, the order of the trials was counterbalanced across participants.

Table 1 describes the pooled mean of reaction rate. It is high at 1.0s, 2.0s, and 4.0s intervals, but low at the intervals of 0.5s. The reaction rate was calculated by the following procedure: (1) The period between a switching of ankle TES and the next was defined as a block. (2) The reaction rate was calculated by dividing the number of blocks where the participant reported a body tilt sensation for the same direction that the system presented by the total number of blocks in a trial. Figure 3 illustrates the excerpts from the participants' body tilt sensation data for each switching interval condition. At 1.0s, 2.0s, and 4.0s intervals, the system's output (black lines) are followed by the user's responses (lines of other colors), while not at the interval of 0.5s. These results indicate that users can perceive the body tilt sensation under the switching intervals of 1.0s, 2.0s, and 4.0s. However, most users can not perceive it under the switching intervals of 0.5s. We have confirmed that switching at 1.0 s or slower intervals elicits a body tilt sensation. Utilizing this finding, we have developed a demonstration of ElectricalVRsKi that does not necessitate switching at intervals shorter than 1.0 s.

4 RESEARCH PLAN

In this paper, we proposed a sensory presentation method that employs dynamically switching ankle TES. We investigated the inter-

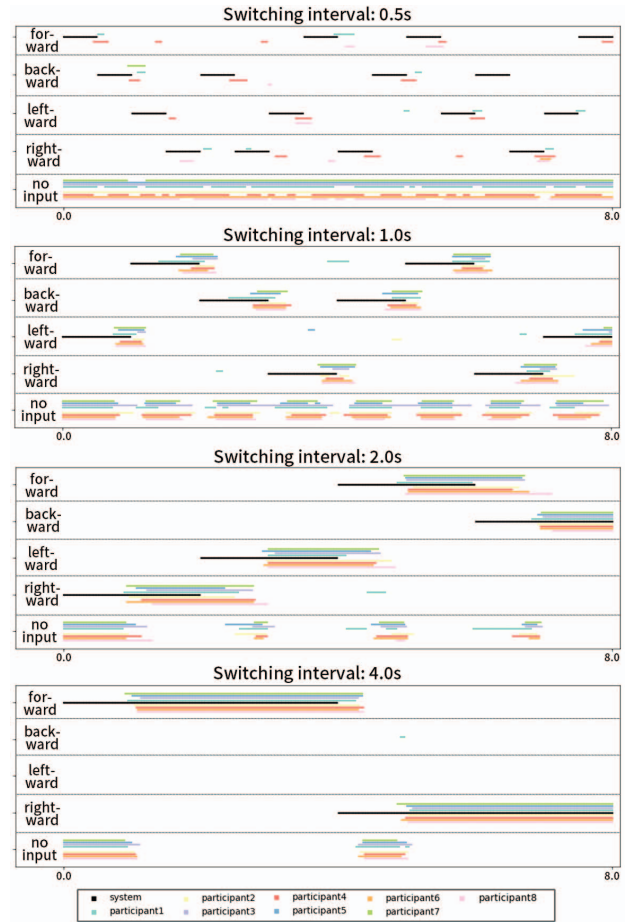


Figure 3: The excerpts of the participants' body tilt sensation data for each switching interval condition. The horizontal axis represents time [s] and the vertical axis represents the direction of body tilt sensation.

vals for switching ankle TES at which the body tilt sensation can be perceived. Furthermore, based on the findings, we developed a VR skiing application. This study suggests that ankle TES has the potential for VR systems that require dynamic sensory presentation. We are currently working on applying ankle TES to VR locomotion systems such as Walking-in-Place and Redirected Walking.

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